Intro to Machine Learning
Quiz 4 Practice Problems
Practice problems, not turned in

Directions: These practice problems cover the material for Quiz 4, which is an open-note in-class worksheet: cubic regression splines in Part 1 and smoothing splines in Part 2. They are not turned in, but the worksheet draws on them, so work through each one before looking at the solutions. You can bring them, with your solutions, to the worksheet. Problems adapted from ISL cite the exercise; see the book for the original wording.

Throughout, $h(x, \xi) = (x - \xi)^3_+$, which is $(x - \xi)^3$ for $x > \xi$ and 0 otherwise (ISL 7.10), and degrees of freedom (df) count every coefficient, including the intercept.

Part 1: Cubic Regression Splines

1. **The truncated power basis by hand.** (ISL 7.9, Exercise 1; ISL 7.4.3.)

   (a) Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x, \xi)$. Show that f, f' and f'' are continuous at ξ, and find the jump in f''' at ξ.

   (b) A cubic spline with one knot at $\xi = 1$ was fitted, giving $\hat{f}(x) = 2 - x + x^2 + 3h(x, 1)$. Compute $\hat{f}(0)$ and $\hat{f}(2)$, and write $\hat{f}$ as a single cubic for $x > 1$.

   (c) A spline with knots at 0 and 1 is $\hat{f}(x) = x - h(x, 0) + 2h(x, 1)$. Compute $\hat{f}(-1)$, $\hat{f}(1)$ and $\hat{f}(2)$. At each knot, by how much does $\hat{f}'''$ jump?

2. **Counting knots and degrees of freedom.** (ISL 7.4.1–7.4.3.)

   (a) You can afford df = 8. How many interior knots does a cubic spline get? A natural cubic spline?

   (b) With $K = 2$ knots, a piecewise cubic has $4 \times 3 = 12$ df. How many does it have if only f is required to be continuous at the knots? If f and f' are?

   (c) In one or two sentences, explain why a natural spline is usually preferred to a cubic spline with the same df.

3. **Training RSS and nested fits.** (ISL 7.4.) True or false? Give a one-sentence reason for each.

   (a) Adding a knot to a cubic regression spline, and keeping the others, never increases the training RSS.

   (b) Adding a knot to a cubic regression spline can increase the test error.

   (c) Moving one knot to a new position, without changing the number of knots, never increases the training RSS.

(d) With boundary knots at the smallest and largest x_i, a natural cubic spline has a training RSS at least as large as the cubic spline with the same interior knots.

Part 2: Smoothing Splines

Here $\hat{g}_\lambda$ minimizes $\sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int g''(t)^2\,dt$ (ISL 7.11), $\mathbf{S}_\lambda$ is its smoother matrix, $\mathrm{df}_\lambda = \sum_i \{\mathbf{S}_\lambda\}_{ii}$ (ISL 7.13).

4. **Fit against roughness.** (ISL 7.5.1, eq. 7.11.) The x_i all lie in $[0, 1]$, and for this problem take the penalty integral over $[0, 1]$. Three candidate curves have these training RSS values:

Curve	$g(x)$	Training RSS
g_1	$1 + 2x$	2.0
g_2	$1 - x + x^2$	0.8
g_3	$1 - x + x^3$	0.2

   (a) Compute the penalty $\int_0^1 g''(t)^2\,dt$ for each curve.

   (b) Compute the criterion $\mathrm{RSS} + \lambda \int_0^1 g''(t)^2\,dt$ for each curve at $\lambda = 0.01$, 0.1 and 1. Which curve has the smallest criterion at each λ?

   (c) Describe in one sentence how the winner changes as λ grows, and why.

   (d) None of the three is the smoothing spline $\hat{g}_\lambda$. At $\lambda = 0.1$, what can you say about the criterion value of $\hat{g}_\lambda$ compared with your answers in (b)?

5. **Regression splines and smoothing splines side by side.** (ISL 7.4, 7.5.) Answer each in one or two sentences.

   (a) A cubic regression spline with K knots, fitted by least squares, has fitted values $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$. What is $\mathrm{tr}(\mathbf{H})$? What plays the role of $\mathbf{H}$ for a smoothing spline?

   (b) For a cubic regression spline you choose K and where the knots go. What do you choose for a smoothing spline?

   (c) Can a cubic regression spline have $\mathrm{df} = 6.4$? Can a smoothing spline?

   (d) A cubic regression spline with 2 knots and a smoothing spline with $\mathrm{df}_\lambda = 6$ are fitted to the same data. Both have 6 df. Name one way the two fits differ.

6. **What the smoothing spline is.** (ISL 7.5.1.) Answer each in one or two sentences.

   (a) For $\lambda > 0$, what kind of function is $\hat{g}_\lambda$, and where are its knots?

   (b) Suppose you took a natural cubic spline with a knot at every x_i and fitted it by ordinary least squares, with no penalty. What would the fitted curve do, and why? What stops the smoothing spline from doing the same?

   (c) Nobody tells the smoothing spline to be linear beyond x_1 and x_n. Why is it?

   (d) "A smoothing spline has n knots, so it has n parameters and must overfit." What is wrong with this?

7. **The two ends of λ.** (ISL 7.5.1–7.5.2.) A smoothing spline is fitted to $n = 50$ points with distinct x_i.

(a) What are df_λ and the training RSS at $\lambda = 0$? What is $\mathbf{S}_0$?

(b) True or false: as $\lambda \to \infty$, $\hat{g}_\lambda$ tends to the horizontal line $\bar{y}$. Explain.

(c) Three fits have $\mathrm{df}_\lambda = 4$, 12 and 30. Which has the largest λ? Which has the largest training RSS?

(d) Can a smoothing spline have $\mathrm{df}_\lambda = 1.5$? Why or why not?

8. **LOOCV without refitting.** (ISL 7.5.2 and 5.1.2, eq. 5.2.) For a smoothing spline,

$$\mathrm{RSS}_{cv}(\lambda) = \sum_{i=1}^{n} \left(y_i - \hat{g}_\lambda^{(-i)}(x_i)\right)^2 = \sum_{i=1}^{n} \left(\frac{y_i - \hat{g}_\lambda(x_i)}{1 - \{\mathbf{S}_\lambda\}_{ii}}\right)^2.$$

A smoothing spline is fitted to $n = 5$ points at two values of λ, with these residuals $e_i = y_i - \hat{g}_\lambda(x_i)$ and diagonal entries of $\mathbf{S}_\lambda$:

Fit		$i = 1$	2	3	4	5
A	e_i	0.1	-0.2	0.25	-0.2	0.1
	$\{\mathbf{S}_\lambda\}_{ii}$	0.9	0.8	0.75	0.8	0.9
B	e_i	0.2	-0.3	0.35	-0.3	0.2
	$\{\mathbf{S}_\lambda\}_{ii}$	0.6	0.4	0.3	0.4	0.6

(a) Find df_λ for each fit. Which was fitted with the larger λ?

(b) Compute the training RSS and RSS_{cv} for each fit. Which λ does LOOCV choose?

(c) What does the left-hand side of the formula mean, and why is it the right thing to compare rather than the training RSS?

(d) Why can't the formula be used at $\lambda = 0$?